October, 2026

Joey Lakerdas-Gayle

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Education

Doctor of Philosophy in Pure Mathematics Expected August 2027
University of Waterloo
Advisors: Barbara Csima & Andy Zucker

Master of Mathematics in Pure Mathematics August 2023
University of Waterloo
Advisor: Barbara Csima

Honours Bachelor of Science with High Distinction June 2022
University of Toronto
Specialist in Mathematics and Major in Computer Science, CGPA: 3.7/4.0
Dean's List (Awarded for obtaining a CGPA of 3.5 or higher)

Research Interests

Computability theory, infinite combinatorics, computable structure theory, structural Ramsey theory

Awards and Grants

Ontario Graduate Scholarship, May 2025 - April 2026
Susan and Janos Aczel Graduate Scholarship, May 2025 - April 2026
Queen Elizabeth II Graduate Scholarship, May 2024 - April 2025
Ontario Graduate Scholarship, May 2023 - April 2024

Publications

Isomorphism Spectra and Computably Composite Structures. *Z. Math. Log. Grundlagen Math.*, to appear (2026), pp. 197-210.
DOI: 10.60866/CAM.260.

The lexicographically least square-free word with a given prefix (with Siddharth Berera, Andrés Gómez-Colunga, John López, Mauditra Matin, Daniel Roebuck, Eric Rowland, Noam Scully, and Juliet Whidden). *The Electronic Journal of Combinatorics*, 30 (2023) #P3.11 (43 pages).
DOI: 10.37236/11659.

Preprints

Weakly uniform computable categoricity and function composition. Submitted for publication.

Oscillating subalgebras of the atomless countable Boolean algebra (with Dana Bartošová, David Chodounský, Barbara Csima, Jan Hubička, Matěj Konečný, Stevo Todorcevic, Spencer Unger, and Andy Zucker). Submitted for publication.
https://arxiv.org/abs/2505.22603.

Talks

1. *Relativizing uniform computable categoricity*, AMS Fall 2026 Eastern Meeting Special Session on Computable Structure Theory, George Washington University, October 3, 2026.

2. *Big Ramsey degrees for infinitely constrained binary free amalgamation classes*, The 26th Graduate Student Conference in Logic, University of Illinois Chicago, April 26, 2026.
3. *Uniform computable categoricity and function composition*, UW-Madison Logic Seminar, April 21, 2026.
4. *Uniform computable categoricity and function composition*, SEALS 2026, University of Florida, February 28, 2026.
5. *Computability theory of function composition*, 2025 Canadian Mathematical Society Winter Meeting, Toronto, December 8, 2025.
6. *Higher dimension partition principles in uncountable Hausdorff spaces*, Infinite Structural Ramsey Theory, BIRS, Banff, November 24, 2025.
7. *Isomorphism Spectra and Uniform Computable Categoricity*, AMS Fall 2025 Central Meeting Special Session on Computability, Saint Louis University, October 19, 2025.
8. *Weakly Uniform Computable Categoricity*, University of Waterloo Logic Seminar, October 10, 2025.
9. *Isomorphism Spectra and Computably Composite Structures*, The 25th Graduate Student Conference in Logic, University of Notre Dame, April 27, 2025.
10. *Symmetrically indivisible and elementarily indivisible structures*, McMaster-Waterloo Model Theory Seminar, October 8, 2024.
11. *Isomorphism Spectra and Computably Composite Structures*, University of Waterloo Logic Seminar, March 13, 2024.

I have also spoken about other topics in logic at various seminars at the University of Waterloo.

Service

University of Waterloo Women in Math Directed Reading Program, Winter 2025

- I supervised two undergraduate students learning about infinite combinatorics. I developed the curriculum for the reading program, wrote course notes, planned weekly meetings, and assigned readings and exercises between meetings.

University of Waterloo Computability Theory Learning Seminar organizer, Fall 2025 - Present

McMaster-Waterloo Model Theory Learning Seminar, co-organizer Fall 2024 - Summer 2025

Teaching Experience

Lecturer, University of Waterloo

MATH 127: Calculus 1 for Sciences	F25

Teaching Assistant, University of Waterloo

PMATH 433/733: Model Theory	W25
PMATH 432/632: First Order Logic and Computability	F23, F24, F26
PMATH 330: Introduction to Mathematical Logic	F22, W23, S23, W24, S26
MATH 235: Linear Algebra II for Honours Mathematics	F23
MATH 136: Linear Algebra I for Honours Mathematics	W23
MATH 135: Algebra for Honours Mathematics	F22, F24
MATH 115: Linear Algebra for Engineering	F26

Teaching Assistant, University of Toronto

CSCC24: Principles of Programming Languages	W21, W22, S22
CSCC63: Computability and Computational Complexity	W22
CSCB63: Design and Analysis of Data Structures	W22
MATA67: Discrete Mathematics	F20, F21
MATA22: Linear Algebra I	W21
MATA36: Calculus II	W21
CSCA08: Introduction to Computer Science I	F19, W20, F20

Math 900 - University Mathematics Teaching Techniques Winter 2025
University of Waterloo

- Completed course providing information and practical experience for teaching undergraduate mathematics.
- Teaching opportunities followed by discussion with experienced lecturers.

Senior Instructor, Science Engagement Programs 2017-2021
York University

- Developed and taught math and computer science curriculum in-person and online to groups of students (grades 3 to 12).
- Planned and led trainings for other instructors about teaching math and computer science.
- Instructed and supervised large groups of students in a classroom setting.
- Taught a wide range of topics in math, computer science, biology, chemistry, and physics.

Work Experience

Software Developer, Pharmer Program Software 2020 - 2023

- Designed and developed a web-app.
- Implemented algorithms based on mathematical models.
- Used knowledge in Full Stack web development, SQL, and Javascript.
- Maintained the app and implemented features requested by customers.